

Misc

1. open interval:

$$(a, b) := \{x \in \mathbb{R} : a < x < b\}$$

2. closed interval:

$$[a, b] := \{x \in \mathbb{R} : a \leq x \leq b\}$$

3. Bernoulli inequality:

$$\boxed{\forall n \in \mathbb{N}_0, \forall a \geq -1 \in \mathbb{R} :}$$

$$\boxed{(1 + a)^n \geq 1 + na}$$

4. Geometric summation formula:

$\sum_{k=1}^n q^k$	$1 + q + q^2 + \dots + q^n$	$\frac{1-q^{n+1}}{1-q}$
$\sum_{k=1}^n a^k b^{n-k}$	$b^n + ab^{n-1} + a^2b^{n-2} + \dots + a^n$	$\frac{a^{n+1}-b^{n+1}}{a-b}$

5. Binomial coefficient rules:

$$\binom{n}{k} = \binom{n}{n-k}$$

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}, k > 0$$

6. Binomial formula:

$$\boxed{\forall n \in \mathbb{N}_0, \forall a, b \in \mathbb{C} :}$$

$$\boxed{(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} b^k a^{n-k}}$$

7. inequality between geometric and arithmetic average:

$$\boxed{\forall n \in \mathbb{N}, a_1, a_2, \dots, a_n \in \mathbb{R}_+}$$

$$\sqrt[n]{a_1 \cdot a_2 \cdot \dots \cdot a_n} \leq \frac{1}{n}(a_1 + a_2 + \dots + a_n)$$

8. inequality properties:

$a + c < b + c$	$\Longleftrightarrow$	$a < b$
$(a < b) \wedge (\Theta \leq \Pi)$	$\Longleftrightarrow$	$a + \Theta < b + \Pi$
$(a < b) \wedge (0 < \Theta \leq \Pi)$	$\Longleftrightarrow$	$a \cdot \Theta < b \cdot \Pi$
$a < b$	$\Longleftrightarrow$	$-b < -a$
$0 < a < b$	$\Longleftrightarrow$	$0 < \frac{1}{b} < \frac{1}{a}$

9. Clopen sets:

$$\boxed{\forall A \neq \emptyset \subset \mathbb{R} : \quad A \text{ (not closed)} \not\Rightarrow A \text{ (open)}}$$

$\mathbb{R}, \emptyset$ are both closed and open (clopen)!

Completeness

10. Triangle inequality:

$$\boxed{\begin{array}{c} \forall a, b \in \mathbb{C} \\ |a| - |b| \leq ||a| - |b|| \leq |a + b| \leq |a| + |b| \\ \\ \forall x, y, z, \dots \in \mathbb{C} : \\ |x + y + z + \dots| \leq |x| + |y| + |z| + \dots \end{array}}$$

11. Equality definition:

$$\forall a, b \in \mathbb{C} \\ a = b \Longleftrightarrow \forall \epsilon > 0 \in \mathbb{R}, |a - b| < \epsilon$$

12. Axiom of completeness:

$$\boxed{\begin{array}{c} \forall A \neq \emptyset \subseteq \mathbb{R}, \\ (\exists s \in \mathbb{R}, \forall x \in A, \quad s \geq x) \implies (\exists! X \in \mathbb{R}, X = \sup A) \end{array}} \text{\$ \$}^7 2b4d8$$

13. Nested interval property:

$$\begin{array}{c} \forall n \in \mathbb{N}, \\ I_n = [a_n, b_n] = \{x \in \mathbb{R} : a_n \leq x \leq b_n\}, \\ I_{n+1} \subseteq I_n \\ \\ \bigcap_{n=1}^{\infty} I_n \neq \emptyset \end{array}$$

14. Archimedean property:

$$\begin{array}{l} 1) \forall x \in \mathbb{R}, \exists n \in \mathbb{N} : n > x \\ 2) \forall x \in \mathbb{R}, \exists n \in \mathbb{N}, \frac{1}{n} < x \end{array}$$

15. Density of $\mathbb{Q}$ in $\mathbb{R}$:

$$\begin{array}{c} \forall a, b \in \mathbb{R} : \\ (a < b) \implies (\exists x \in \mathbb{Q}, a < x < b) \end{array}$$

16. Schröder-Bernstein Theorem:

$$\boxed{\begin{array}{l} \exists f : X \rightarrow Y \text{ Injektiv} \\ \exists g : Y \rightarrow X \text{ Injektiv} \end{array}} \implies \exists \phi : X \rightarrow Y \text{ Bijektiv}$$

17. Cantor theorem:

$$\boxed{\begin{array}{l} \forall X, \forall \Psi \in \mathcal{P}(X)^X, \exists \Upsilon \in \mathcal{P}(X) : \\ \forall x \in X, \Psi(x) \neq \Upsilon \end{array}}$$

Sequences

18. Sequence definition:

Sequence / Folge $(a)_n$ in S :
 $a : \mathbb{N} \rightarrow S, n \mapsto a_n$

19. Convergence of a Sequence:

$$\begin{aligned} &\text{sequence } (a)_n \text{ converges to } x \\ &\lim_{n \rightarrow \infty} a_n = x \\ &\iff \\ &\forall \epsilon > 0 \in \mathbb{R}, \exists L \in \mathbb{N} : \\ &\forall n \geq L \in \mathbb{N}. |a_n - x| < \epsilon \end{aligned}$$

20. Supremum analytically:

$$\begin{aligned} &\forall A \subseteq \mathbb{R}, \sup A = X \\ \iff &(\forall a \in A, X \geq a) \wedge (\forall \epsilon > 0 \in \mathbb{R}, \exists a \in A, X - a < \epsilon) \end{aligned}$$

21. Neighborhood in $\mathbb{R}/\mathbb{C}$:

$$V_\epsilon(x) = \{z \in (\mathbb{R}/\mathbb{C}) : |z - x| < \epsilon\}$$

$$\boxed{\epsilon > 0 \in \mathbb{R}}$$

22. Bounded Sequence def. :

$$\begin{aligned} &\text{Bounded sequence } (a)_n : \\ \iff &\exists M \in \mathbb{R}, \forall n \in \mathbb{N}, |a_n| \leq M \end{aligned}$$

23. Algebraic limit theorem:

$$\boxed{\lim a_n = A, \lim b_n = B}$$

1. $\forall c \in \mathbb{R}, \lim(ca_n) = cA$
2. $\lim(a_n + b_n) = A + B$
3. $\lim(a_n b_n) = AB$
4. $\lim\left(\frac{a_n}{b_n}\right) = \frac{A}{B}, B \neq 0$

24. Order limit theorem:

$$\boxed{\lim a_n = A, \lim b_n = B}$$

1. $\forall n \in \mathbb{N}, (a_n \geq 0) \implies A \geq 0$
2. $\forall n \in \mathbb{N}, (a_n \leq b_n) \implies A \leq B$
3. $\forall c \in \mathbb{R}, \forall n \in \mathbb{N}, (a_n \geq c) \implies A \geq c$

25. squeeze theorem:

$$\boxed{\begin{array}{c} \forall n \in \mathbb{N}, a_n \leq b_n \leq c_n \\ \wedge \\ \lim a_n = \lim c_n = X \end{array}} \implies \lim b_n = X$$

26. Monotone def. :

Monotone increasing:

$$\forall n \in \mathbb{N}, a_{n+1} \geq a_n$$

Monotone decreasing:

$$\forall n \in \mathbb{N}, a_{n+1} \leq a_n$$

27. Monotone convergence theorem:

$$\boxed{\begin{array}{c} (a)_n \text{ is bounded} \\ \wedge \\ (a)_n \text{ is monotone} \end{array}} \implies (a)_n \text{ converges}$$

28. Subsequence def. :

$(a)_n$ is a sequence

$(n)_k : \mathbb{N} \rightarrow \mathbb{N}$ is a sequence with $\forall k \in \mathbb{N} : n_{k+1} > n_k$

$(a)_{n_k}$ is subsequence of $(a)_n$

29. Convergence through subsequences:

$$(a)_n \text{ conv.} \iff \text{all subseq. conv.}$$

30. Existence of monotone subsequences:

every real sequence has a monotone subseq.

31. Bolzano-Weierstraß theorem ($\mathbb{R}/\mathbb{C}$):

$$\boxed{\begin{array}{c} \text{Bounded sequence } (a)_n: \\ \iff \exists M \in \mathbb{R}, \forall n \in \mathbb{N}, |a_n| \leq M \\ \implies \\ \boxed{\exists (a)_{n_k} \text{ conv. subseq.}} \end{array}}_{(\mathbb{R}/\mathbb{C})}$$

32. Cauchy sequence:

Cauchy sequence :

$$\forall \epsilon > 0 \in \mathbb{R}, \exists L \in \mathbb{N} : \\ \forall m, n \geq L \implies |a_n - a_m| < \epsilon$$

33. Cauchy criterion:

$$(a)_n \text{ conv. } \iff (a)_n \text{ is Cauchy seq.}$$

34. Completeness implications:

$$\text{AoC} \implies \begin{cases} \text{NIP} \implies \text{BW} \implies \text{CC} \\ \text{MCT} \end{cases}$$

AoC:	Axiom of Completeness
NIP:	Nested Interval Property
MCT:	Monotone Convergence Theorem
BW:	Bolzano-Weierstraß Theorem
CC:	Cauchy Convergence Criterion

Complex numbers

35. Set of complex numbers:

$$\mathbb{C} := \{z = x + iy : \forall x, y \in \mathbb{R}\} \\ i := +\sqrt{-1}$$

$$\text{Re}(z) = x$$

$$\text{Im}(z) = y$$

$$\bar{z} := x - iy \text{ complex conjugate}$$

36. Properties of complex conj. and abs. value:

$$\forall z, w \in \mathbb{C} :$$

$$\text{Re}(z) = \frac{1}{2}(z + \bar{z}), \text{Im}(z) = \frac{1}{2i}(z - \bar{z})$$

$$\overline{z + w} = \bar{z} + \bar{w}$$

$$\overline{z \cdot w} = \bar{z} \cdot \bar{w}$$

$$\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}, w \neq 0$$

$$\overline{\bar{z}} = z$$

$$z \cdot \bar{z} = |z|^2 = \text{Re}(z)^2 + \text{Im}(z)^2 \geq 0$$

$$z = \bar{z} \iff z \in \mathbb{R}$$

37. absolute complex properties:

$$\forall z, w \in \mathbb{C} : \\ |\operatorname{Re}(z)| \leq |z|, |\operatorname{Im}(z)| \leq |z|, \\ |\operatorname{Re}(z)| + |\operatorname{Im}(z)| \geq |z|$$

$$|z| = 0 \iff z = 0 \\ |z \cdot w| = |z| \cdot |w|, \left| \frac{z}{w} \right| = \frac{|z|}{|w|} \\ \frac{1}{z} = \frac{\bar{z}}{|z|^2}$$

triangle inequality:

$$|z + w| \leq |z| + |w| \\ ||z| - |w|| \leq |z - w|$$

Series

38. Series definition:

series is a sequence: :

$$s_n : n \mapsto \sum_{i=1}^n a_i$$

$(a)_i$ is a sequence of $\mathbb{R}/\mathbb{Q}$

$$\boxed{\sum_{i=1}^{\infty} a_i = A \iff \lim s_n = A}$$

39. Algebraic limit theorem for series:

$$\boxed{\sum_{i=1}^{\infty} a_i = A, \sum_{i=1}^{\infty} b_i = B}$$

$$1) \sum_{i=1}^{\infty} c \cdot a_i = c \cdot A \\ 2) \sum_{i=1}^{\infty} (a_i + b_i) = A + B$$

40. Cauchy criterion for series:

$$\sum_{i=1}^{\infty} a_i \text{ conv.} \\ \iff \forall \epsilon > 0 \in \mathbb{R}, \exists L \in \mathbb{N} : \\ \forall n > m \geq L \in \mathbb{N} : \\ |s_n - s_m| < \epsilon$$

41. Convergence and zero asymptote:

$$\sum_{i=1}^{\infty} a_i \text{ conv.} \implies \lim a_i = 0$$

42. Series comparison test:

$$\boxed{\begin{array}{l} (a)_n, (b)_n \text{ are seq. with} \\ \forall n \in \mathbb{N} : 0 \leq a_n \leq b_n \end{array}}$$
$$\implies \begin{cases} \sum_{n=1}^{\infty} b_n \text{ conv.} \implies \sum_{n=1}^{\infty} a_n \text{ conv.} \\ \sum_{n=1}^{\infty} a_n \text{ diver.} \implies \sum_{n=1}^{\infty} b_n \text{ diver.} \end{cases}$$

43. Absolute convergence test:

$$\sum_{i=1}^{\infty} |a_i| \text{ conv.} \implies \sum_{i=1}^{\infty} a_i \text{ conv.}$$

44. Alternating series test:

$$\begin{array}{l} (a)_n \text{ seq. with:} \\ \bullet a_{n+1} \leq a_n \\ \bullet \lim(a)_n = 0 \end{array}$$
$$\implies \sum_{n=1}^{\infty} (-1)^{n+1} a_n \text{ conv.}$$

45. Absolute vs. conditional convergence:

$$\boxed{\text{Absolute Convergence:}}$$

$$\sum_{i=1}^{\infty} |a_i| \text{ conv.}$$

$$\boxed{\text{Conditional Convergence:}}$$

$$\sum_{i=1}^{\infty} a_i \text{ conv.} \wedge \sum_{i=1}^{\infty} |a_i| \text{ diver.}$$

46. Rearrangements definition:

$$\begin{array}{l} \text{rearrangement of } \sum_{i=1}^{\infty} a_i : \\ f \in \mathbb{N}^{\mathbb{N}} \text{ (bijektiv)} \end{array}$$

$$\sum_{i=1}^{\infty} a_{f(i)} = \sum_{i=1}^{\infty} b_i$$

$$\boxed{\text{called a rearrangement of sum}}$$

47. absolute convergence implies rearrangement convergence:

$$\sum_{i=1}^{\infty} |a_i| = A$$
$$\implies \forall f(\text{bijektiv}) \in \mathbb{N}^{\mathbb{N}} : \sum_{i=1}^{\infty} a_{f(i)} = A$$

48. Summation by parts:

$(a)_i, (b)_i$ seq. with:

$$a_0 = b_0 = 0$$

$$\Delta a_i = a_{i+1} - a_i$$

$$\Delta b_i = b_{i+1} - b_i$$

$$\sum_{i=m}^n b_i \cdot \Delta a_i = (a_{n+1}b_{n+1} - a_mb_m) - \sum_{i=m}^n \Delta b_i \cdot a_{i+1}$$

49. Root test for series:

$$\sum_{i=1}^{\infty} a_k \text{ conv.} \iff \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$$

50. Abel test :

$$\left[\begin{array}{c} \sum_{i=1}^{\infty} a_i \text{ conv.} \\ \wedge \\ (b)_i : \forall i \in \mathbb{N}, b_{i+1} \leq b_i \end{array} \right] \implies \left[\begin{array}{c} \sum_{i=1}^{\infty} a_i \cdot b_i \\ \text{is conv.} \end{array} \right]$$

51. Harmonic series:

$$S = \sum_{x=1}^{\infty} \frac{1}{x} = \infty$$

$$S_n \approx \ln(n) + \gamma$$

$$\gamma = 0.57 \dots$$

52. Cauchy condensation test:

$$\forall n \in \mathbb{N}, (a_n \geq 0) \wedge (a_{n+1} \leq a_n) :$$

$$a_n \text{ conv.} \iff \sum_{n=0}^{\infty} 2^n a_{2^n} \text{ conv.}$$

53. Dirichlet test:

$$\left[\begin{array}{c} \exists L \in \mathbb{R}, \forall n \in \mathbb{N}, \sum_{i=1}^n a_i = s_n \leq L \\ \text{(bounded)} \\ \wedge \\ (b)_i : \forall i \in \mathbb{N}, b_{i+1} \leq b_i \wedge \lim(b)_i = 0 \end{array} \right] \implies \left[\begin{array}{c} \sum_{i=1}^{\infty} a_i \cdot b_i \\ \text{is convergent} \end{array} \right]$$

Topology

54. Open set def. :

$$O \subseteq \mathbb{R} \text{ is Open:}$$

$$\forall x \in O, \exists \epsilon > 0 \in \mathbb{R} :$$

$$V_\epsilon := (x - \epsilon, x + \epsilon) \subseteq O$$

55. accumulation / limiting points:

$$A \subseteq \mathbb{R} \text{ has limiting point } x \text{ if:}$$

$$\exists (a)_i : \mathbb{N} \rightarrow A \setminus \{x\} :$$

$$\lim (a)_i = x$$

otherwise $x \in A$ is called isolated point

56. closed set def. :

$$\boxed{A \subseteq \mathbb{R} \text{ is closed}}$$

$$\iff$$

$$\mathcal{H}_A \subseteq A$$

$$\mathcal{H}_A = \{a : a \text{ is limiting point of } A\}$$

57. Cauchy condition for closed sets:

$$A \subseteq \mathbb{R}$$

$$A \text{ is closed}$$

$$\iff$$

every Cauchy seq. on A converges

58. Closure def. and properties:

$$\text{Closure of } A \subseteq \mathbb{R} :$$

$$A \cup \mathcal{H}_A = \overline{A}$$

$$\overline{\overline{A}} = \overline{A}$$

Closure is closed

59. complement of sets:

$$\forall A \subseteq \mathbb{R} :$$

$$A \text{ closed} \iff A^C \text{ open}$$

$$A \text{ open} \iff A^C \text{ closed}$$

$\emptyset, \mathbb{R}$ are both open and closed

60. union and intersection of open / closed sets:

$$\forall i \in \mathbb{N}$$

A_i Open:

$$\bigcap^n A_i \text{ open}$$

$$\bigcup A_i \text{ open}$$

A_i Closed:

$$\bigcap A_i \text{ closed}$$

$$\bigcup^n A_i \text{ closed}$$

61. Open cover definition:

A is a Set

Cover:	$\{U_\alpha\}_{\alpha \in I}$	$A \subseteq \bigcup_{\alpha \in I} U_\alpha$
Open Cover:	$\forall U_\alpha \in \{U_\alpha\}_{\alpha \in I}, U_\alpha \text{ is Open}$	
finite Subcover:	$\exists \mathcal{F} \subseteq I, \mathcal{F} < \aleph_0 :$	$A \subseteq \bigcup_{\alpha \in \mathcal{F}} U_\alpha$

62. Compact definitions and Heine-Borel:

A Compact : Every Open cover has finite subcover for A

Heine-Borel Theorem:	Compact
A is Compact	$\Longleftrightarrow$
A Closed and bounded	$\Longleftrightarrow$
All sequences in A have conv. subseq.	$\Longleftrightarrow$
Every open cover of A has a finite subcover	$\Longleftrightarrow$

63. Perfect Set definition:

$A \subseteq \mathbb{R}$ is perfect $\Longleftrightarrow A$ is closed with no isolate points

$$\forall A \subseteq \mathbb{R}, A \text{ is perfect, } |A| = \aleph_1$$

64. Separated, disconnected, and connected Set definitions :

A, B are separated $\Longleftrightarrow$	$A, B \neq \emptyset \subset \mathbb{R} :$	$(A \cap \overline{B} = \emptyset) \wedge (\overline{A} \cap B = \emptyset)$
E is disconnected $\Longleftrightarrow$	$E \subset \mathbb{R}, \exists A, B \neq \emptyset \subset E, E = A \cup B :$	$(A \cap \overline{B} = \emptyset) \wedge (\overline{A} \cap B = \emptyset)$
E is connecteed $\Longleftrightarrow$	$E \subset \mathbb{R}, \forall A, B \neq \emptyset \subset E, E = A \cup B :$	$(A \cap \overline{B} \neq \emptyset) \vee (\overline{A} \cap B \neq \emptyset)$
E is connected $\Longleftrightarrow$	$E \subset \mathbb{R}, \forall A, B \neq \emptyset \subset E, E = A \cup B :$	$(\exists (a)_n \in A^\mathbb{N} \vee \exists (a)_n \in B^\mathbb{N}) \wedge ((\lim (a)_n) \in E)$
E is connected $\Longleftrightarrow$	$\forall a, b \in E :$	$(a < c < b) \implies c \in E$

Continuity

65. Definition of functional limit at x_0 :

$$f : A \rightarrow \mathbb{R}, x_0 \text{ is limiting point of } A$$

$$\lim_{x \rightarrow x_0} f(x) = L$$

$$\forall \epsilon > 0, \exists \delta > 0, \forall x \in A : |x - x_0| < \delta \implies |f(x) - L| < \epsilon$$

66. Limits mean convergence of every sequence in A :

$$f : A \rightarrow \mathbb{R}, x_0 \text{ is limiting point of } A$$

$$\begin{array}{c} \lim_{x \rightarrow x_0} f(x) = L \\ \iff \\ \text{Every seq. } (a)_n, a_n \neq x_0 \text{ in } A \\ \text{and } \lim(a)_n = x_0 : \lim f(a_n) = L \end{array}$$

67. Functional limits laws:

$$f, g : A \rightarrow \mathbb{R}, A \subseteq \mathbb{R}$$

$$\lim_{x \rightarrow x_0} f(x) = L \quad \lim_{x \rightarrow x_0} g(x) = M$$

$\lim_{x \rightarrow x_0} k \cdot f(x)$	$=$	$k \cdot L$
$\lim_{x \rightarrow x_0} f(x) + g(x)$	$=$	$L + M$
$\lim_{x \rightarrow x_0} f(x) \cdot g(x)$	$=$	$L \cdot M$
$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)}$	$=$	$\frac{L}{M}, \forall x \in A, g(x) \neq 0$

68. Functional Squeeze Theorem:

$$\begin{array}{c} \forall x \in A \subseteq \mathbb{R} \\ f(x) < g(x) < h(x) \end{array}$$

$$\lim_{x \rightarrow x_0} f(x) = L \quad \lim_{x \rightarrow x_0} h(x) = L$$

$$\implies \lim_{x \rightarrow x_0} g(x) = L$$

69. Continuous without Limit explicitly:

$$f : A \rightarrow \mathbb{R}, x_0 \text{ is limiting point of } A$$

$$\forall \epsilon > 0, \exists \delta > 0, \forall x \in A :$$

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \epsilon$$

70. Continuity Topologically, and sequentially:

$f : A \rightarrow \mathbb{R}, x_0$ Limiting point of $A, A \subseteq \mathbb{R}$	Continuity:
f is Continuous at x_0	$\Longleftrightarrow$
$\forall \epsilon > 0, \exists \delta > 0, \forall x \in A : x - x_0 < \delta \implies f(x) - f(x_0) < \epsilon$	$\Longleftrightarrow$
$\forall \epsilon > 0, \exists \delta > 0, \forall x \in A : x \in V_\delta(x_0) \implies f(x) \in V_\epsilon(f(x_0))$	$\Longleftrightarrow$
$\forall (a)_n \in A^{\mathbb{N}}, \lim(a)_n = x_0 \implies \lim f((a)_n) = f(x_0)$	$\Longleftrightarrow$
x_0 is Limit point of $A \implies \lim_{x \rightarrow x_0} f(x) = f(x_0)$	$\Longleftrightarrow$

71. Continuity preserving Operations:

f, g are Continuous

$\implies$

$k \cdot f(x)$
$f(x) + g(x)$
$f(x) \cdot g(x)$
$\frac{f(x)}{g(x)}, g(x) \neq 0$
$f \circ g$
Continuous

72. Continuity and preimage condition:

$f : A \rightarrow \mathbb{R}$

f is continuous

$\Longleftrightarrow \forall B \text{ (Open)} \subseteq \mathbb{R}, f^{-1}(B) = A \cap C(\text{Open})$

73. continuous compact image is continuous compact:

$f : A \rightarrow \mathbb{R}$ is Continuous $X \subseteq A$ is Compact $\implies f(X)$ is Compact

74. Extreme value theorem:

$\forall f(\text{Contin.}) : A \rightarrow \mathbb{R}, X \subseteq A(\text{Compact}) \implies \exists a, b \in f(X) : \sup(f(X)) = a, \inf(f(X)) = b$

75. Intermediate Value theorem:

$\forall f(\text{Contin.}) : [a, b] \rightarrow \mathbb{R}, f(a) < f(b) : f(a) < L < f(b) \implies \exists x \in (a, b), f(x) = L$

76. Preservation of Connected sets:

$\forall f(\text{Contin.}) : A \rightarrow \mathbb{R}, C \subseteq A$ is connected $\implies f(C)$ is connected

77. Strictly monotone and surjective implies existence of inverse:

$f(\text{Contin.}) : A \rightarrow B, A, B \subseteq \mathbb{R}$
 $\forall x, y \in A, x < y \implies f(x) < f(y)$ (Mono.) $\implies \exists f^{-1}(\text{Contin.}) : B \rightarrow A$ (Bij.)
 $\forall z \in B, \exists w \in A : f(w) = z$ (Surj.)

